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Equipment transport bag

Abstract

Equipment transport bag assemblies (20, 92, 98, 145, 152, 182) each including an upright, open-top, equipment-receiving bag (22, 110, 158) with associated equipment-handling assemblies (24, 92, 102, 164, 184). The assemblies (24, 92, 102, 164, 184) have load-bearing strap units (34, 114, 166, 186) operably associated with the transport bags (22, 110, 158) so that the strap units (34, 114, 166, 186) bear the primary loads during lifting and handling of the bag assemblies (20, 92, 98, 145, 152, 182).

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of U.S. patent application Ser. No. 18/054,605, filed Nov. 11, 2022, which is a continuation of U.S. patent application Ser. No. 16/026,942, filed Jul. 3, 2018, the entire disclosure of each which is hereby incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention is concerned with equipment containment bag assemblies designed to provide environmentally safe shipping and storage conditions for leaking or otherwise non-performing equipment, such as pole-mounted distribution transformers, switchgear, breakers, and reclosers, equipped with lifting appendages (e.g., lifting lugs, eyes, or hooks). More particularly, the invention is concerned with such bag assemblies having an open-top equipment-receiving bag with an equipment-handling assembly associated with the bag and including at least one load-bearing strap unit designed to engage the at least one lifting appendage, and interconnected bag-supporting sections. In this way, the load-bearing strap units bear the primary loads associated with lifting and handling of equipment within the bag.

Description of the Prior Art

(2) Conventional electrical transformers used in power distribution systems include a sealed, oil-filled tank with internal electrical components, such as transformer coils. These transformers also

have external hardware, such as mounting equipment, connection busses, and lifting lugs. These transformers are subject to leaking over time, resulting from damage to the transformer tank or from extended use. Such leakages can be environmentally damaging, inasmuch as the oil formulations within the transformers can often include hazardous chemicals (e.g., PCBs). In any event, it is necessary to safely transport failed transformers for disposal or repair, in accordance with governmental regulations. In like manner, other types of equipment, and especially electrical utility equipment, require safe and effective containment and shipping enclosures.

(3) A number of different transformer containment devices have been provided in the past. For example, Andax Industries LLC has commercialized Xtra HD Pole-Mount Transformer Sac™ containment bags, which have large, puncture-proof transformer bags with opposed perforated sections which can be opened to allow access to transformer lifting lugs. However, these bags do not provide any closures for these openings, and accordingly moisture or contaminants can enter the bags.

(4) U.S. Pat. No. 8,777,001 describes another type of transformer containment bag which includes a bag with an internal liner and external lifting hoops. These kinds of bags are deficient, and indeed do not meet relevant regulatory standards, because the lifting lugs of the transformers cannot be accessed, thus requiring that the transformers be elevated and moved using only the bag components, which places considerable stress on the overall assemblies.

SUMMARY OF THE INVENTION

(5) The present invention overcomes the problems outlined above and provides equipment transport bag assemblies, which allow lifting and handling of equipment having one or more lifting appendages without imposing undue loads upon the equipment bags. To this end, the bag assemblies of the invention comprise an upright bag (which may be of any desired shape or size, such as circular or quadrate) including a bottom wall and upwardly extending sidewall structure, the bag presenting an open upper end and configured to receive equipment therein; an equipment-handling assembly is operably associated with the bag and equipment and includes at least one load-bearing strap unit for operably engaging the at least one lifting appendage, with associated structure engaging and supporting the bag so that the equipment is retained within the bag during handling thereof.

(6) In certain embodiments where the equipment includes at least a pair of spaced lifting lugs, the equipment-handling assembly includes a pair of load-bearing strap units each having a first strap section with an arcuate portion configured to engage a corresponding lifting lug, and a second strap section with an arcuate lifting portion, the first and second strap sections operably connected together. Again, the equipment-handling assembly is operably coupled with the bag so that the equipment is retained within the bag during handling thereof. As used herein, “lifting lugs” or “lugs” refer to any type of integrated lifting point(s) associated with the equipment to allow safe lifting and handling thereof. The term “operably connected together” with respect to the strap sections and arcuate lifting sections refers to the fact that these sections may be configured using individual straps which are then interconnected, or from a single strap.

(7) To this end, each of the first lug-engaging strap sections may be formed from a first strap of material having opposed ends, the opposed ends secured to the sidewall structure by stitching or any other suitable technique. Likewise, the second strap sections are formed from a second strap of material having opposed ends, the opposed ends of the second straps secured to the sidewall structure and to the first straps. The load-bearing strap sections may be formed as reverse loops or as bight sections.

(8) In other embodiments where the equipment lifting appendage(s) comprise one or more lifting eyes, typically mounted at the top central region of the equipment. In such cases, the equipment-handling assembly is operably connected with the equipment and the bag and includes at least one load-bearing strap unit operably coupled with the lifting appendage; other structure may be used to secure the bag, such as a separate strap passing through the lifting eye.

(9) Advantageously, in all of the embodiments, the equipment-handling assembly is designed so that the strap units bear a substantial portion of the lifting load, normally at least about 85% of the load of the equipment, and usually essentially 100% thereof. In this way, the bag is not unduly stressed during handling of the equipment, but rather primarily serves as a containment vessel, and bears only the load of its weight and that of any collected fluid therein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of one embodiment of an equipment transport bag in accordance with the invention, shown with an electrical transformer within the bag, ready for lifting;
- (2) FIG. 2 is a fragmentary vertical sectional view illustrating one of the load-bearing strap units of the transport bag of FIG. 1;
- (3) FIG. 3 is a perspective view similar to that of FIG. 1, but illustrating a lift assembly operably coupled with the equipment transport bag;
- (4) FIG. 4 is a vertical sectional view of the FIG. 1 embodiment, through the bag adjacent the central load-bearing straps, shown with the transformer supported by the load-bearing strap units;
- (5) FIG. 5 is a perspective view of another equipment transport bag embodiment in accordance with the invention, illustrated with an electrical transformer therein;
- (6) FIG. 6 is a view similar to that of FIG. 5, but depicting a lift assembly operably coupled with the transport bag;
- (7) FIG. 7 is a perspective view of another equipment transport bag embodiment in accordance with the invention of quadrate configuration, illustrated with an electrical transformer therein;
- (8) FIG. 8 is a view similar to that of FIG. 7, but depicting a lift assembly operably coupled with the transport bag;
- (9) FIG. 9 is a perspective view of another equipment transport bag embodiment in accordance with the invention of quadrate configuration, illustrated with an electrical transformer therein;
- (10) FIG. 10 is a view similar to that of FIG. 9, but depicting a lift assembly operably coupled with the transport bag;
- (11) FIG. 11 is a vertical sectional view of the FIG. 7 embodiment, through the bag adjacent the central load-bearing straps, shown with the transformer supported by the load-bearing strap units;
- (12) FIG. 12 is a perspective view of another embodiment wherein the lifting appendage is in the form of a central lifting eye, and wherein the equipment handling assembly comprises a pair of strap units; and
- (13) FIG. 13 is another embodiment wherein the lifting appendage is in the form of a central lifting eye, and wherein the equipment handling assembly comprises a single strap having a first portion engaging the lifting eye, second lifting portions respectively passing through openings formed in the bag sidewall, and third lifting portions.
- (14) While the drawings do not necessarily provide exact dimensions or tolerances for the illustrated components or structures, FIGS. 1-13 are to scale with respect to the relationships between the components of the structures illustrated therein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Embodiments of FIGS. 1-11

- (15) These embodiments are particularly designed for use with equipment such as transformers, which are universally provided with a pair of opposed, side-mounted lifting lugs.
- (16) Turning now to FIGS. 1-4, an equipment transport bag assembly 20 is depicted, which broadly includes an upright, open-top, generally circular in plan bag 22, together with an equipment-handling assembly 24. As illustrated, an electrical transformer 26 is received within bag 22, and has a pair of opposed, outwardly-extending lifting lugs 28 (FIGS. 2 and 4).

(17) The bag **22** includes a bottom wall **30** and upwardly extending sidewall structure **32** terminating in an upper margin **33**. The bag **22** may be fabricated from any suitable material, but is preferably formed of heavy-duty synthetic resin material, and is substantially leak-proof. As explained below, the function of bag **22** is primarily to contain any spillage of liquid from the transformer **26**, and does not function as the primary load-bearing component during the lifting and handling of the transformer **26**.

(18) The assembly **24** includes a pair of opposed, load-bearing strap units **34**, which, in this embodiment, are secured to the sidewall structure **32**. Each such load-bearing strap unit **34** has a first section **36** with an arcuate portion **38** designed to engage a corresponding lifting lug **28**, and a second strap section **40** with an arcuate lifting portion **42**.

(19) Each first strap section **36** is formed from an elongated first strap **44** having first and second ends **46** and **48**. As illustrated, the ends **46**, **48** terminate adjacent the upper margin **33** of sidewall structure **32**, with the portion **38** being a loop **39** intermediate these ends. Each second strap section **40** is formed using a second, generally U-shaped strap **50** having opposed ends **52** and **54**. The strap **50** presents a central bight **56** and upwardly extending, opposed legs **58**. As shown, the bight **50** extends across bottom wall **30**, whereas the legs **58** extend upwardly along the sidewall structure **32** to the upper margin **33**; from this point, the legs **58** extend further upwardly and then downwardly to form each lifting portion **42** in the form of a loop **59**. The bight **56** and legs **58** are secured to the bottom wall **30** and sidewall structure **32**, respectively, by stitching or any other appropriate means.

(20) The first and second strap sections **36** and **40** are secured to each other, and to sidewall structure **32** adjacent the upper margin **33**. As illustrated in FIG. 2, the upper end of sidewall structure **32** is sandwiched between the ends **46**, **48** of the first sections **36**, and to the legs **58** and the terminal portions of the lifting portions **42**. Such interconnections are preferably accomplished by means of stitching, but could be achieved by other means, such as by heat welding or suitable adhesive.

(21) The overall equipment-handling assembly **24** also includes a pair of opposed, non-load-bearing, bag-orienting strap sections **60** and **62**, each having a corresponding arcuate connection portion **64**, **66**. In this case, the strap sections **60**, **62** are formed using a third U-shaped strap **68** presenting a central bight (not shown) and upstanding legs **70**. The bight and legs **70** are secured to the bottom **30** of bag **22** and to the sidewall structure **32** thereof, respectively. The upper ends of each of the legs **70** extend above the margin **33** and are reversed to form loops **72**. Again, the end portion of each loop **72** is secured to sidewall structure **32** by stitching or other appropriate means.

(22) In order to provide further reinforcement for the bag **22**, a pair of circumferentially extending upper and lower straps **74**, **76** extend around sidewall structure **32** and the legs **58**, and are secured thereto.

(23) In order to effectively lift and transport the transformer **26** within the bag **22**, an exemplary lift assembly **78** is employed (FIGS. 3-4). The assembly **78** includes a fourth strap **80** having endmost loops **82**, as well as an identical fifth strap **84** with endmost loops **86**. A clevis slip hook **88** is secured to each of the loops **82** and **86**. As shown, the slip hooks **88** of strap **80** are secured to the loops **59**, whereas the slip hooks **88** of strap **84** are secured to the loops **72**. A single hook **90** engages the straps **80** and **84** at the central regions thereof, and is connected to a further lifting arrangement (not shown) to allow lifting and handling of bag **22** and the transformer **26** therein.

(24) During such lifting and manipulation of bag **22**, it will be appreciated that, owing to the construction of the strap units **34**, these units and the associated lift assembly **78** bear a substantial part of the lifting load, and that the third strap **68**, and the bag **22** itself, bear only a minor portion or none of the lifting load. In effect, the strap **68** and the associated loops **72** function primarily as a means of controlling the bag **22** during lifting and handling operations. For example, as set forth in FIG. 4, it will be observed that during lifting the transformer **26** is simply contained within bag **22** and does not engage the bag **22** at all. Thus, the strap units bear essentially all of the load of the

equipment, while the bag **22** bears its own weight and any fluid collected therein.

(25) FIGS. **5-6** illustrate another embodiment in accordance with the invention, which is similar in many respects to the embodiment of FIGS. **1-4**. Accordingly, identical parts will be given the same reference numerals. The principal difference between the embodiment of FIGS. **5-6** and that of FIGS. **1-4** lies in the fact that, instead of the reverse loops **59** and **72**, arcuate sections **94** and **96** are used. Thus, the second U-shaped strap **50a** has legs **58a** which extend upwardly along and are connected to the sidewall structure **32**, but extend further upwardly and laterally to form the arcuate bight sections **94** and then extend downwardly along the sidewall structure **32** in spaced relationship to the legs **58a**. Similarly, the upwardly extending legs **70a** of third U-shaped strap **68a** extend further above the margin **33** and then laterally to form bight sections **96**, with the terminal portions of the strap **68a** extending downwardly and connecting to sidewall structure **32** in spaced relationship to the legs **70a**. In all other respects, the embodiment of FIGS. **5-6** is identical to that of FIGS. **1-4**.

(26) FIGS. **7-8** depict a third embodiment in the form of a transport bag assembly **98** including an upright, open-top, substantially quadrate in plan bag **100**, together with an equipment-handling assembly **102**. Equipment **104** is received within bag **100**, and is equipped with side-mounted lifting lugs **106**.

(27) The bag **100** includes a bottom wall (not shown) together with sidewall structure **108** in the form of four interconnected sidewall panels **110** cooperatively defining an upper margin **112**. The bag **100** is flexible and is formed of appropriate synthetic resin or other suitable material.

(28) The handling assembly **102** includes a pair of opposed, load-bearing strap units **114** secured to sidewall structure **108**. Each such strap unit **114** has a first section **116** designed to engage a corresponding lifting lug **106**, and a second strap section **118** with an arcuate lifting portion **120**. Each of the strap sections **116** are very similar to the sections **36** described above, and include an elongated strap **122** having a loop **124** with the strap ends **126** secured to the inner face of sidewall structure **108** adjacent margin **112**. Each strap section **118** is made up of an elongated strap **128** secured to the outer face of sidewall structure **108**, with an extension above margin **112** in the form of a reverse loop **130**. The strap sections **116** and **118** are secured to each other, and to bag sidewall structure **108** by stitching or any other appropriate means.

(29) The handling assembly **102** also includes a total of four substantially non-load-bearing, bag-orienting strap sections **132** located at the corners of the bag. Each strap section **132** includes an elongated strap **134** secured to the exterior face of sidewall structure **108** and extending above margin **112** with a reverse loop **136**. The end of loop section **136** is secured to the upper end of strap **134**, and to sidewall structure **108**.

(30) A lift assembly **138** is provided for lifting and handling of the bag **100** with equipment **104** therein. The assembly **138** is very similar to that of previously described assembly **78**, and includes a lift strap **140** having slip hooks **88** secured to the ends thereof and coupled with the loops **130**. Additionally, two other straps **142** and **144** are provided, each having endmost slip hooks **88**, and designed to couple with the corner loops **136**.

(31) FIGS. **9-11** illustrate a still further embodiment of the invention in the form of a bag assembly **145**, which is very similar to that of FIGS. **7-8**, and again, like components are identified by the same reference numerals. The only difference between the embodiment of FIGS. **9-11** versus that of FIGS. **7-8** lies in the non-load-bearing corner straps. In FIGS. **9-10**, a continuous, generally U-shaped strap **146** is provided adjacent each corner of the bag **100**. The straps **146** include an uppermost bight **148** and two downwardly extending legs **150**. The latter are secured to the outer faces of the adjacent sidewall panels **110**, as illustrated. Likewise, the lift assembly **138** differs only in the connection of the straps **142** and **144** to the bights **148**, rather than the loops **136**.

(32) Just as in the earlier embodiments, the strap units **114** serve as the primary load-bearing structure, so that the equipment **104** is separate from the bag **100** when elevated.

(33) Each of the embodiments illustrated in FIGS. **1-11** make use of load-bearing strap units having

first and second strap sections which are interconnected and secured to the equipment-receiving bag. It should be understood, however, that the invention is not limited to these types of constructions. For example, the load-bearing strap units can be formed from a single strap, which may be connected to the bag or pass through a slit or other opening in the bag sidewall, without any direct connection between the bag and the strap units. In the latter case, the bag is lifted with the single strap units and is retained in place by the latter and the bag-orienting strap sections.

Embodiments of FIGS. 12 and 13

(34) These embodiments are especially designed for use with equipment having a central, top-mounted lifting eye, in lieu of or in addition to side-mounted lifting lugs.

(35) FIG. 12 illustrates an equipment transport bag assembly **152**, which is useful in connection with equipment **154** having an upstanding, central lifting eye **156**. As in the case of the earlier embodiments, the assembly **152** has a bag **158**, here equipped with a pair of opposed, reinforced opening slits **160**, **162** in the sidewall of the bag. The assembly **152** also has an equipment-handling assembly **164** in the form of a first generally U-shaped load-bearing strap unit **166**, which has a lower segment **168** engaging the eye **156**, and upper loop sections **170** each designed to receive a slip hook **88**. The overall assembly **164** also includes a bag-supporting strap **172** having a central section **174** passing through the eye **156**, with outboard side loop sections **176**, **178** passing through the corresponding opening slits **160**, **162**. One end of the strap **172** is equipped with a metallic ring **180**, whereas the other end has opposed Velcro sections allowing passage through ring **180** and connection to itself.

(36) As will be appreciated from this structure, essentially all of the lifting load is borne by the first strap **166**, with the strap **172** supporting only the weight of the bag **158**.

(37) FIG. 13 depicts another equipment transport bag assembly **182** to be used with the equipment **154** having lifting eye **156**. The assembly **182** is made up of the bag **158** as well as a modified equipment-handling assembly **184**. The latter has a single strap unit **186** having terminal loops **188** and **190**. As shown, the strap unit has a central section **192** which passes through eye **156**, as well as outboard loops sections **194** and **196** respectively passing through the bag opening slits **160**, **162**. Finally, the strap **186** has third lifting portions in the form of the loops **188**, **190**. Both of these loops are designed to interfit with a slip hook **88**. Again, it will be appreciated that essentially all of the lifting load is carried by the assembly **184**, and little if any of such load is transferred to the bag **158**.

Claims

1. An apparatus for use with an equipment lift assembly, said apparatus comprising: equipment including a lifting element; and an equipment transport bag assembly handling the equipment, said bag assembly including an upright bag defining an interior receiving the equipment therein, said bag including an opening slit, said bag assembly further including an equipment-handling assembly, said equipment-handling assembly including a strap unit, said strap unit including a first section located at least in part within the bag interior, with the first section engaging the lifting element of the equipment located within the bag interior, said strap unit including a second section extending through the opening slit, said equipment-handling assembly including a lifting portion located at least in part outside the bag interior and configured to operably couple with the equipment lift assembly, said lifting portion operably coupled to the second section of the strap unit, such that the equipment-handling assembly supports the equipment within the bag interior and functions as the primary load-bearing component of the equipment transport bag assembly during lifting and handling of the equipment using the equipment transport bag assembly.
2. The apparatus of claim 1, said opening slit being reinforced.
3. The apparatus of claim 2, said bag including a bottom wall and sidewall structure extending upwardly from the bottom wall, said opening slit being formed in the sidewall structure.

4. The apparatus of claim 1, said lifting portion and the second section of the strap unit being integrally formed by a single strap of material.
 5. The apparatus of claim 1, said bag including a pair of opening slits, said strap unit including a pair of second sections, each extending through a corresponding one of the opening slits.
 6. The apparatus of claim 5, said opening slits being diametrically opposed from one another.
 7. The apparatus of claim 5, said equipment-handling assembly including a pair of lifting portions, each being operably coupled to a corresponding one of the second sections and each being configured to operably couple with the equipment lift assembly.
 8. The apparatus of claim 7, each of said lifting portions and the corresponding one of the second sections being integrally formed by a single strap of material.
 9. The apparatus of claim 8, said first section, second sections, and lifting portions being integrally formed by the single strap of material.
 10. The apparatus of claim 8, said opening slits being reinforced.
 11. The apparatus of claim 8, said bag including a bottom wall and sidewall structure extending upwardly from the bottom wall, said opening slits being formed in the sidewall structure.
 12. The apparatus of claim 11, said opening slits being diametrically opposed from one another.
 13. The apparatus of claim 5, said first section and said second sections being integrally formed by a single strap of material.
 14. The apparatus of claim 13, said equipment-handling assembly including a pair of lifting portions, each being operably coupled to a corresponding one of the second sections and each being configured to operably couple with the equipment lift assembly.
 15. The apparatus of claim 14, said lifting portions being integrally formed by a second single strap of material.
 16. The apparatus of claim 15, each of said straps of material engaging the lifting element of when the equipment located within the bag interior.
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